DVCon India 2026 - Stage 2A Report

Intelligent Object Selection for Assistive Robotics: LLM-Driven Detection on RISC-V Edge Deployment

Shreelakshmi SP, Pradeep P | RTL Rebels, Team 616 | VIT Vellore

Overview

This pipeline detects objects in the scene and performs reasoning based on semantic knowledge distilled from a large language model to determine which object would be suitable for the given task. In contrast to existing object detection frameworks which treat all objects in the same way, our pipeline incorporates semantic reasoning capabilities.

Stage 1 — Object Detection

YOLOv8-nano operates on CPU and takes the input image as an input. It detects all objects present in the image along with their bounding boxes, class names, and confidence scores. This step provides us with the candidates which are further passed to the semantic reasoning step.

Stage 2 — LLM-Distilled Knowledge Bank

All 80 object classes in COCO and 14 tasks were described semantically by the help of ChatGPT and stored offline. For instance, "A couch refers to soft cushioned sofa with armrests, high comfort level, excellent back support; best suited for relaxed sitting position". On the other hand, "A potted plant refers to decorative purpose only, not used for seating, food, beverage consumption". Semantic descriptions of the objects and tasks were embedded using Sentence-BERT into 384-dimensional dense vectors and stored offline as pre-computed knowledge bank.

Step 3 – Cosine Similarity Scoring

Cosine similarity is calculated for the object's knowledge bank embedding vector and the task query embedding vector. As a result, a number from 0 to 1 representing the object's relevance to the given task is determined. A relevance threshold of 0.25 is set – any objects with a score lower than that value will not be considered relevant due to being a bad choice if there is nothing relevant in the scene.

Step 4 – Weighted Fusion and Selection

In order to generate an aggregate score for each object, semantic similarity is weighted at 70%, and YOLO confidence at 30%. The objects are then ranked according to their scores, and the one above the threshold is chosen as the selection result. The chosen object is marked with a green bounding box on the image alongside its score.

Deliverables based on Stage2a

- Used YOLOv8-nano in place of YOLO-tiny for Stage 2A software implementation using the CPU
- Knowledge bank now utilizes real Sentence-BERT embeddings based on ChatGPT descriptions
- Set relevance threshold at 0.25 for handling scenes with irrelevant objects only
- Incorporated a weighted score from semantic score and detection confidence

Results

Detected Objects from the image using YOLOv8-nano:



For Query “sit comfortable”:



For Query “read”:



For Query “cut something”: FALSE Case

```
Query: cut something

Matching results:
vase => 0.23
book => 0.14
book => 0.14
couch => -0.002
chair => -0.062

No suitable object found for task: cut something
Reason: No detected object scored above threshold of 0.25
Best available: vase | Score: 0.23
Tip: Use an image that contains objects relevant to this task.
```